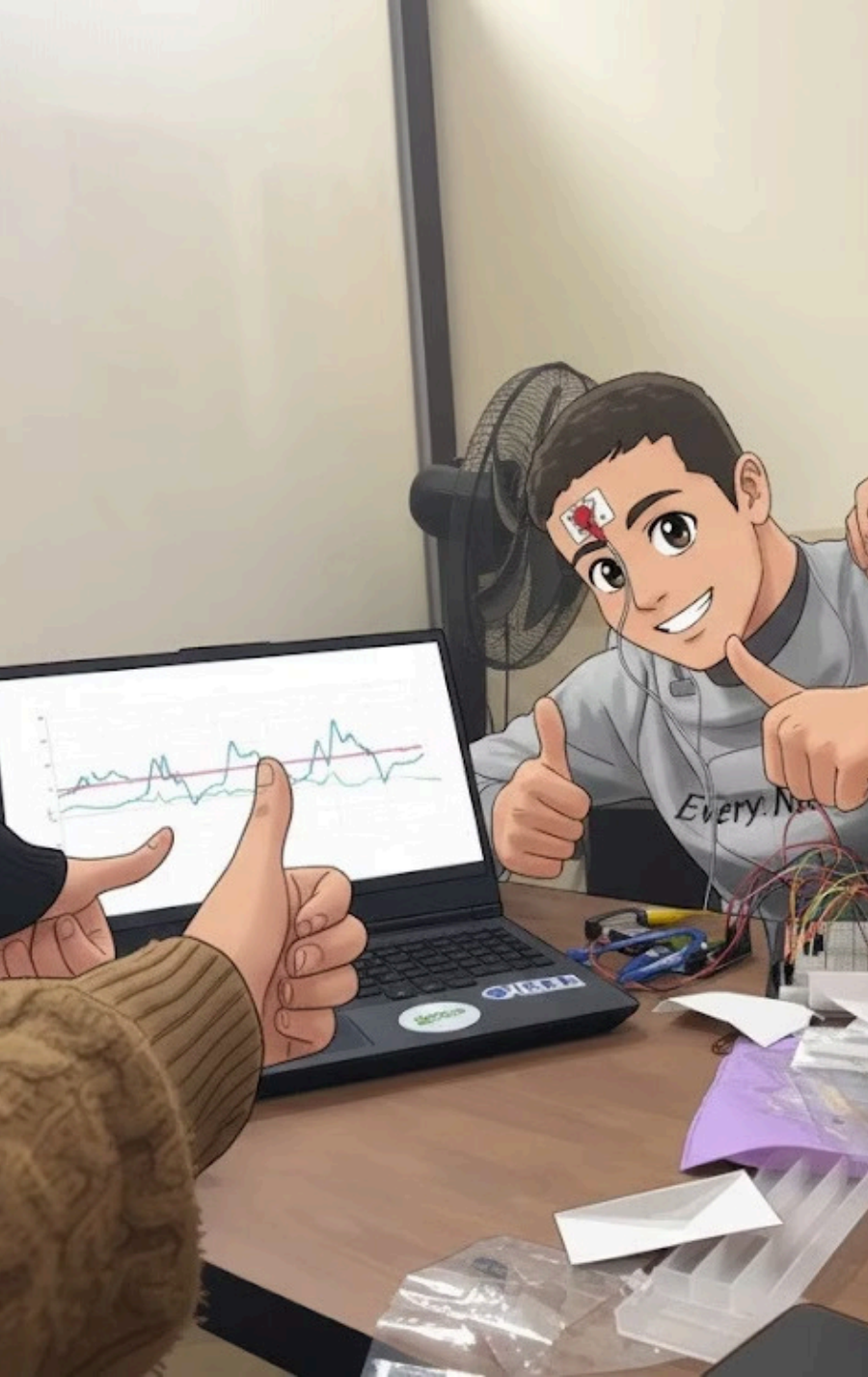




Electroencephalography (EEG) System Design and Signal Analysis Project

Biomedical Engineering Department

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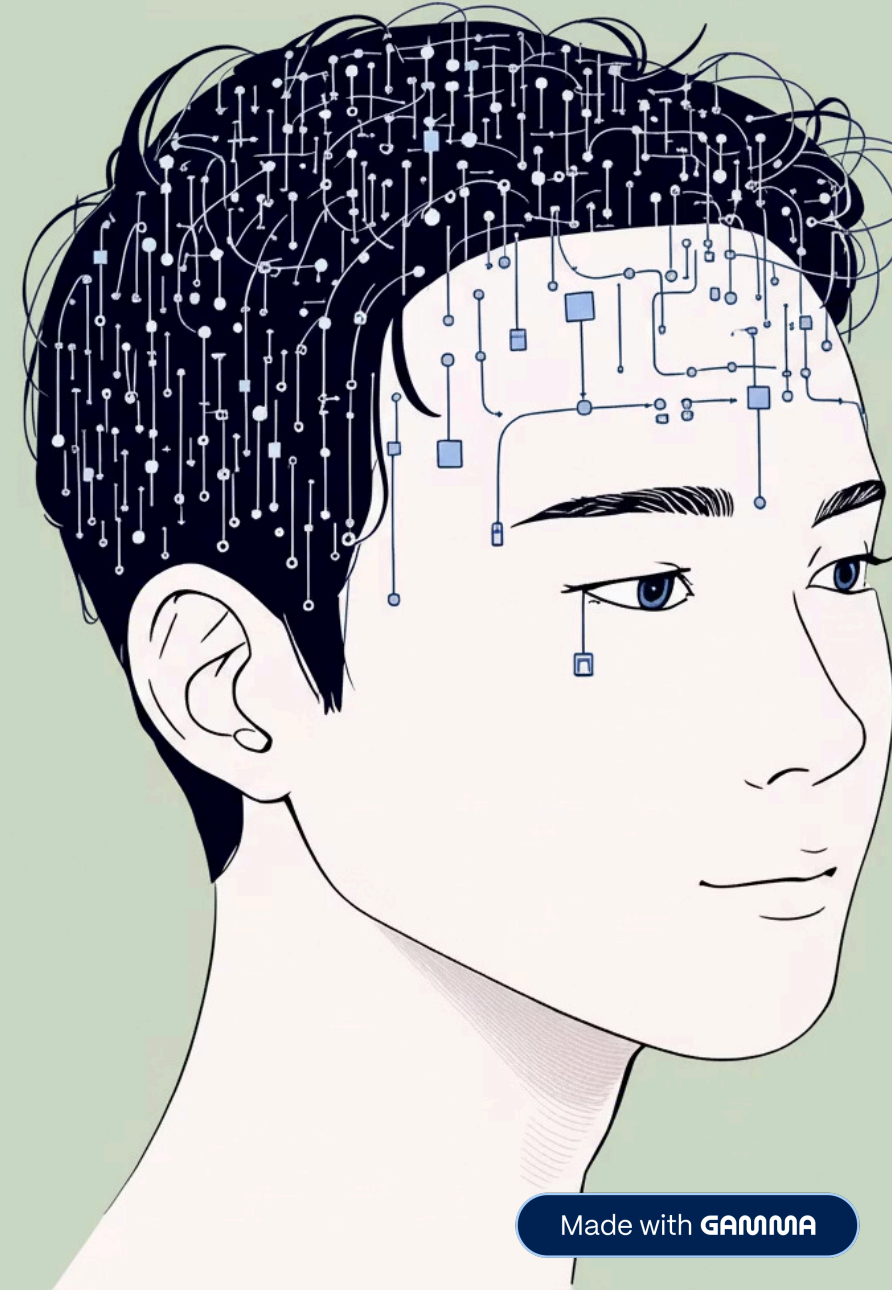
Instructor:

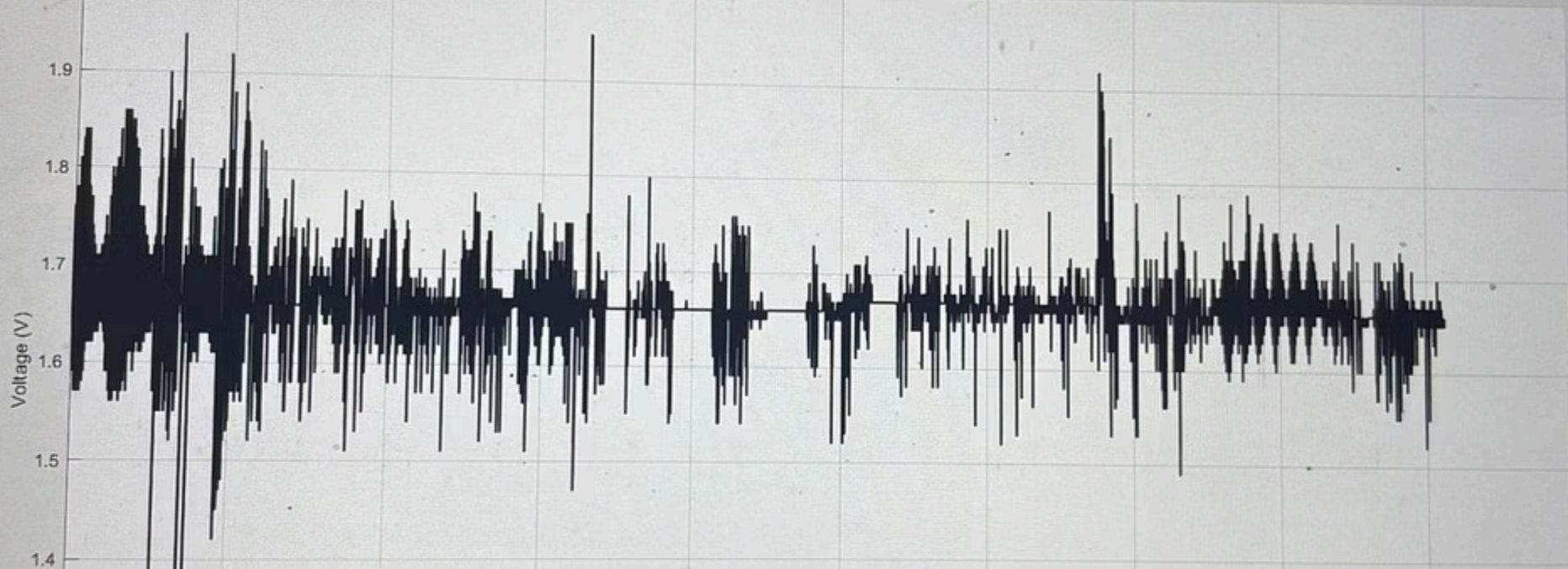
Dr/ Ashraf Mahrous

What is EEG:

EEG stands for Electroencephalograph

- It records the electrical activity of brain
- During an EEG test , small electrodes like cup or disc type are placed on the scalp.
- They pick up the brain's Electrical signals and send them to a machine called Electroencephalograph





EEG Signal Characteristics: A Delicate Balance

EEG signals are subtle and complex, requiring precise measurement, amplification, and rigorous filtering:

- **Frequency Range:** 0.5 - 100 Hz, capturing diverse brain activity.
- **Signal Amplitude:** Very low (2 - 200 μV), prone to interference.
- **Noise Sensitivity:** Sensitive due to low amplitude, demanding advanced signal processing

The Language of the Brain: EEG Wave Types

Brain states are defined by distinct EEG frequency bands:

Delta Waves

(0.5-4 Hz): Deep sleep; associated with restorative activity.

Theta Waves

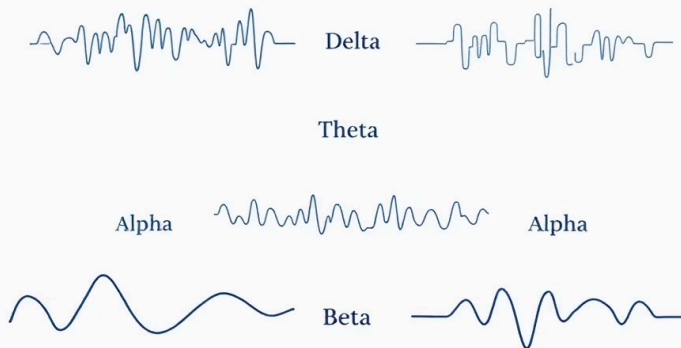
(4-8 Hz): Drowsiness, meditation, light sleep; linked to memory.

Alpha Waves

(9-17 Hz): Relaxed, wakeful (eyes closed); indicates calm mind.

Beta Waves

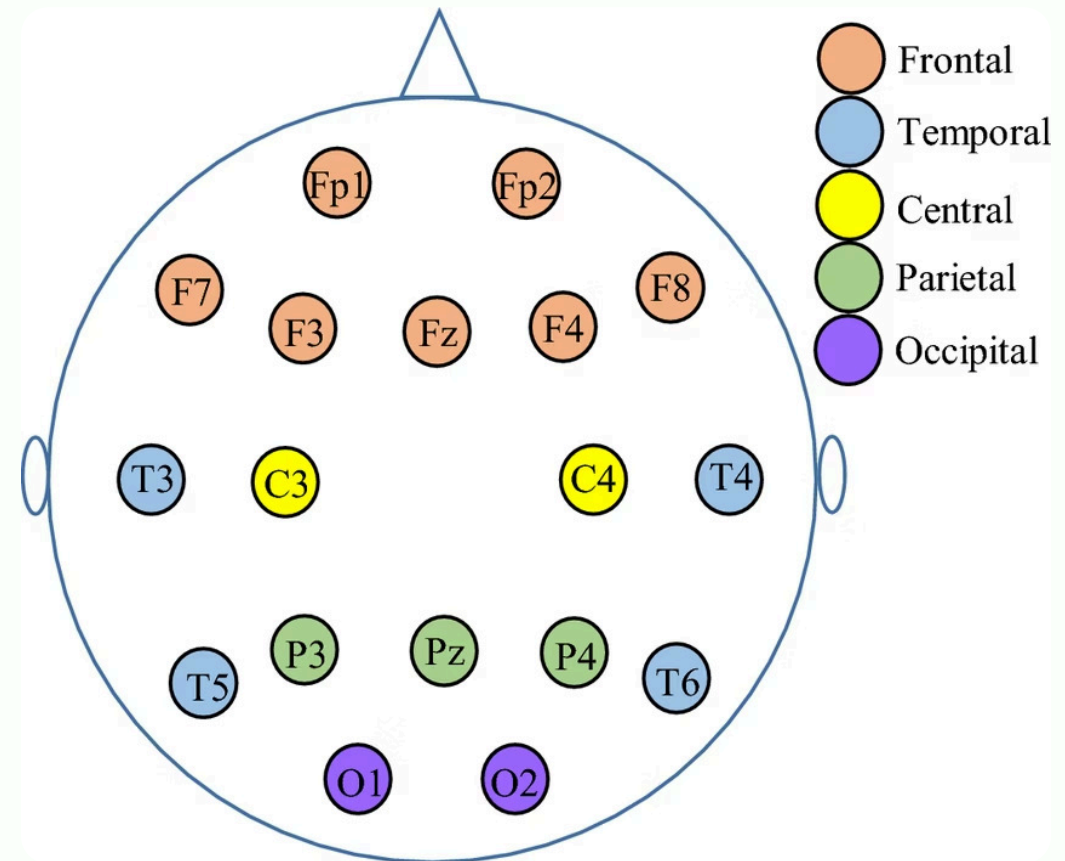
(18-25 Hz): Active thinking, problem-solving, focused mental activity, alertness.



Standardizing Measurement: The 10–20 Electrode System

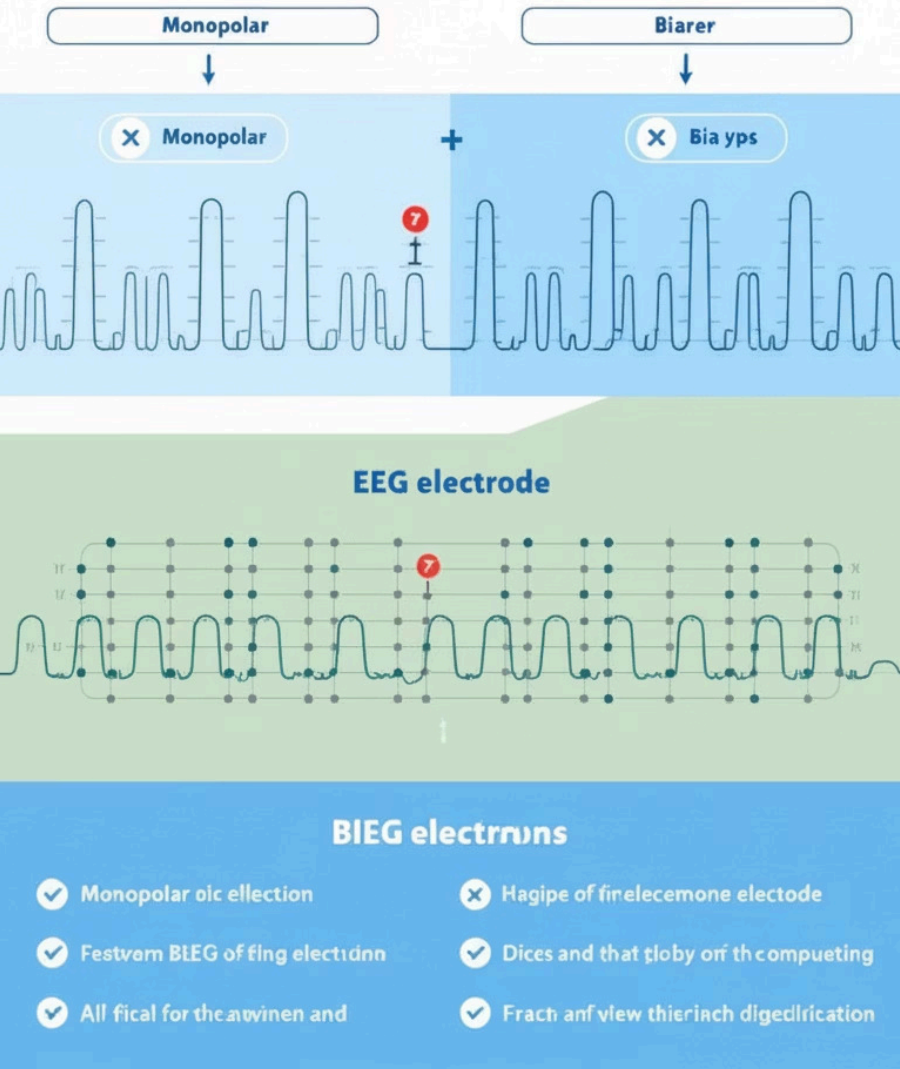
The International 10–20 System standardizes EEG electrode placement.

It uses anatomical landmarks and proportional distances to ensure precise, reproducible positioning, crucial for consistent and comparable measurements across studies.



Monopolar vs EEG electrodes

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your acoduction of loud conctction adies in thar cantlor for taat in that monopolar.



EEG Recording Strategies: Monopolar vs. Bipolar

Recording strategy significantly impacts EEG signal interpretation.

Monopolar Recording

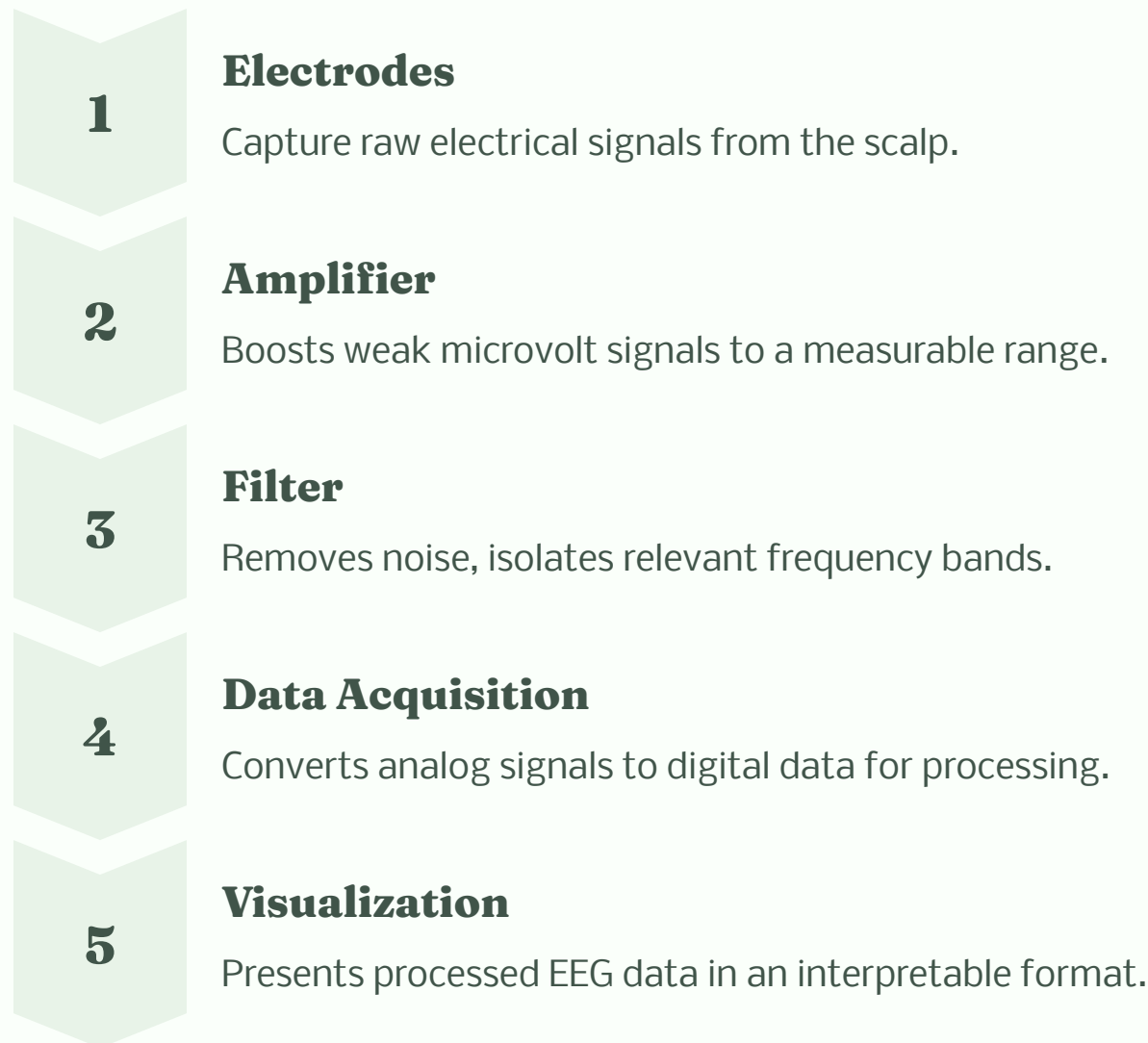
Measures potential between an active electrode and a distant, inactive reference (e.g., mastoid). Offers a broad view of brain activity.

Bipolar Recording

Measures potential between two adjacent active electrodes. Highlights localized activity and reduces common-mode noise.

EEG System Architecture: A Block Diagram Overview

Our EEG system follows sequential stages for signal acquisition and processing:

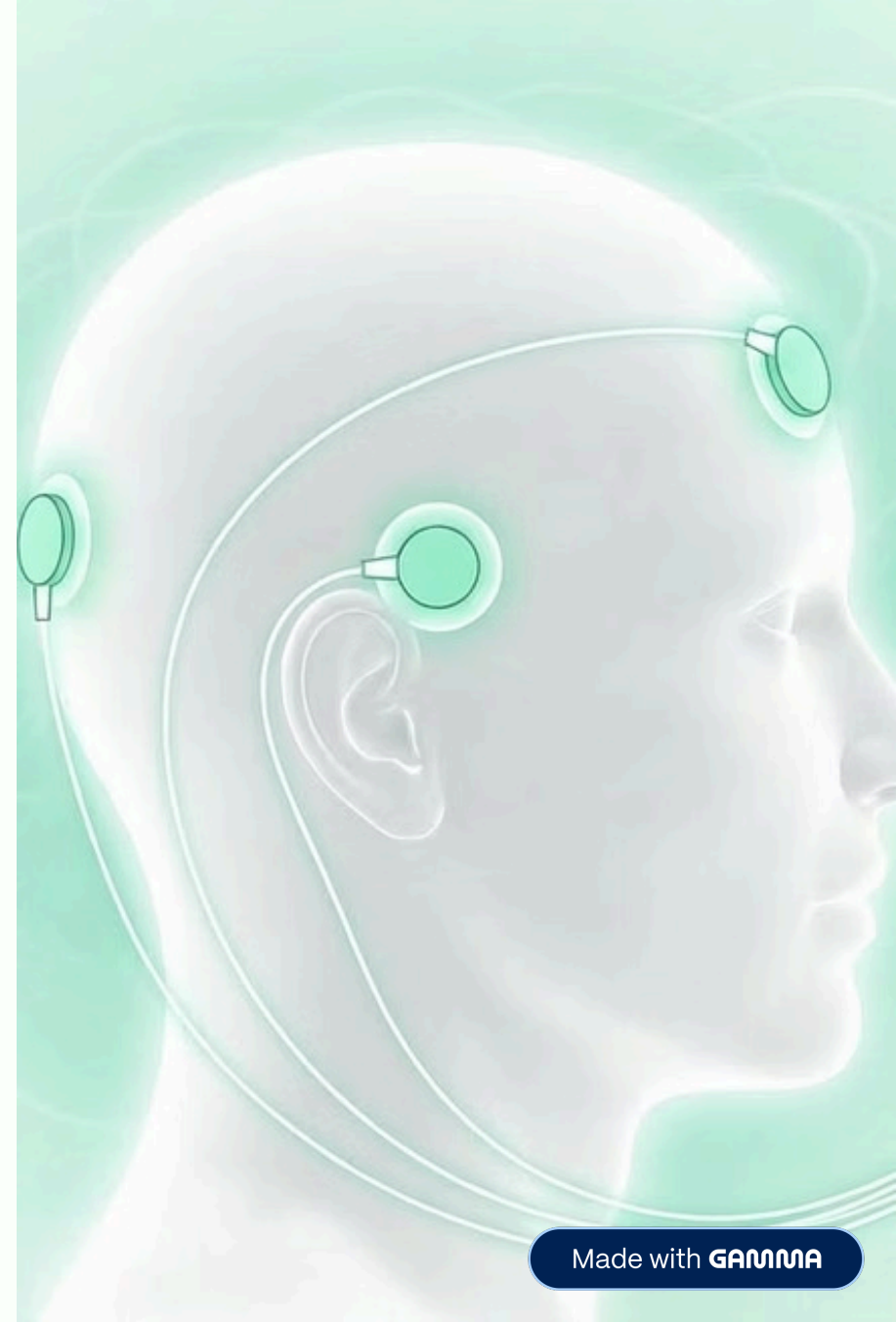


Each stage ensures high fidelity of brain signals, from acquisition to analysis.

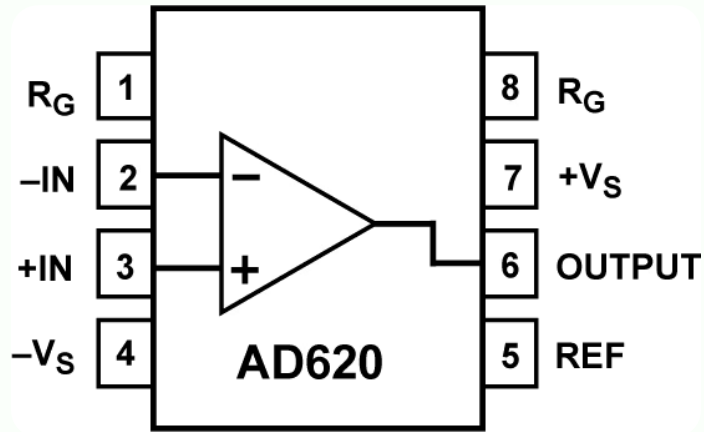
Electrodes Placement: Key Locations

Accurate and consistent electrode placement is fundamental for reliable EEG recordings, ensuring the capture of specific brain activity and minimizing interference. Our setup focuses on key anatomical regions:

1. **Reference Electrode:** Positioned behind the right ear, this electrode serves as a stable baseline, against which the electrical potential at other sites is measured.
2. **Posterior Electrode:** Placed on the posterior region of the scalp, it targets the occipital area, primarily to capture brain activity related to visual processing.
3. **Frontal Electrode:** Located on the frontal region of the scalp, this electrode is designed to record activity associated with the frontal lobe, which is involved in executive functions, memory, and emotional regulation.

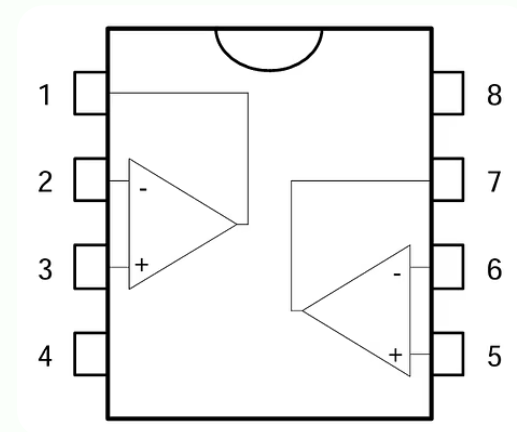


Amplifiers



Instrumentation Amplifier "AD620"

Provide an initial amplification of the EEG signal while rejecting Common-mode interference.



Operational Amplifier "TL072"

High input impedance, Low-Noise Characteristics and Suitability for Bio-signal Amplification

EEG Signal Processing: Refining Brain Data

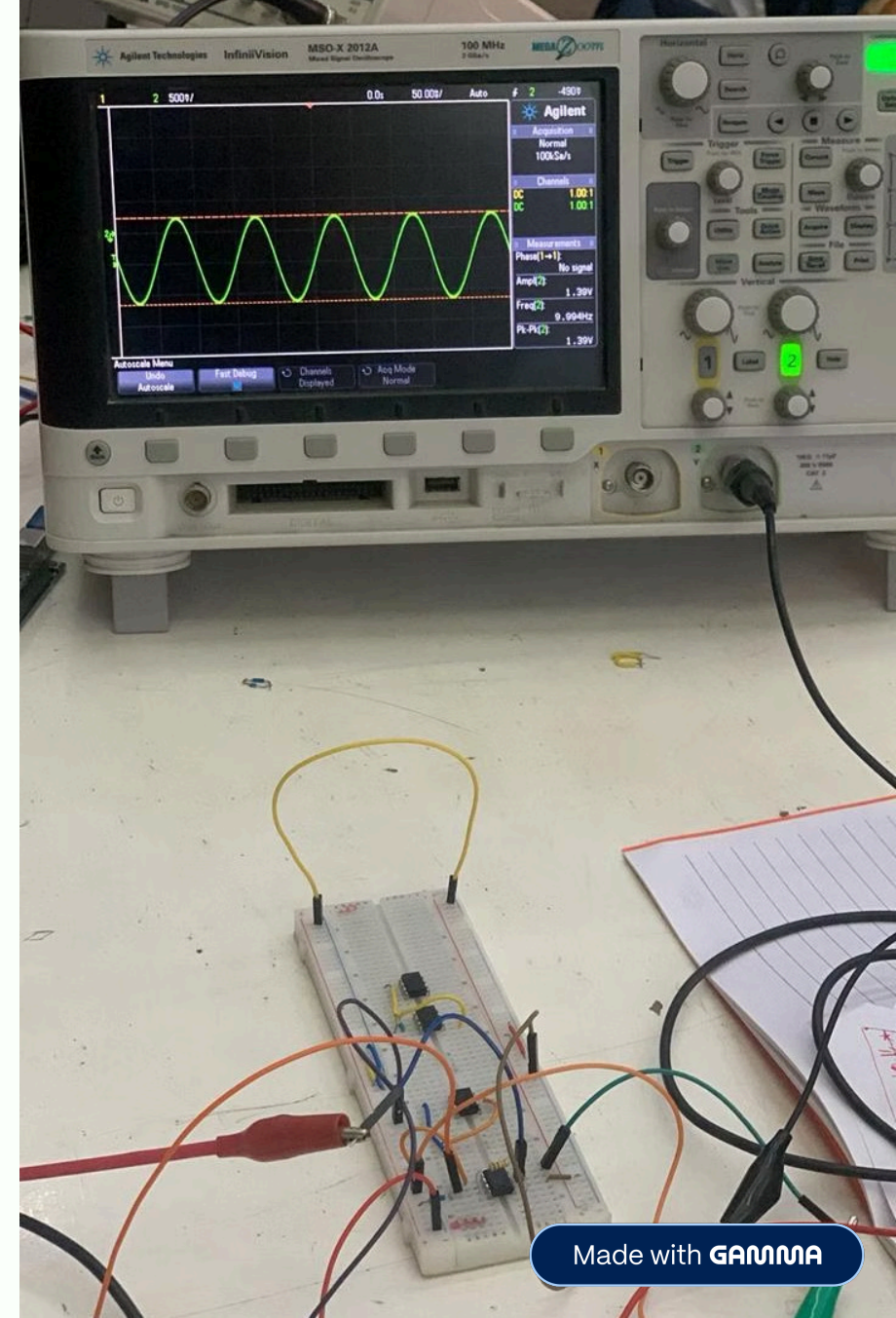
EEG signal processing is critical for isolating genuine brain activity from noise. This multi-stage process enhances signal quality, making it suitable for accurate analysis and interpretation

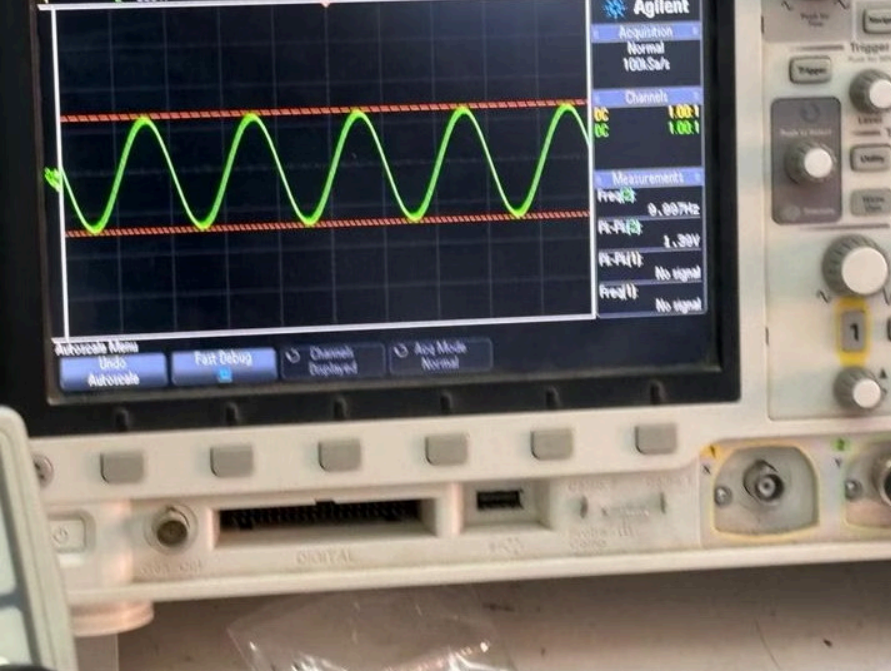
Instrumentation Amplifier (IA)

The IA helped us to:

- Amplify the differential EEG signal between electrodes
- Reject common-mode noise such as power-line interference
- Improve the signal-to-noise ratio (SNR)

This made the recorded EEG signal more stable and reliable.





EEG Signal Noise Reduction and Filtering

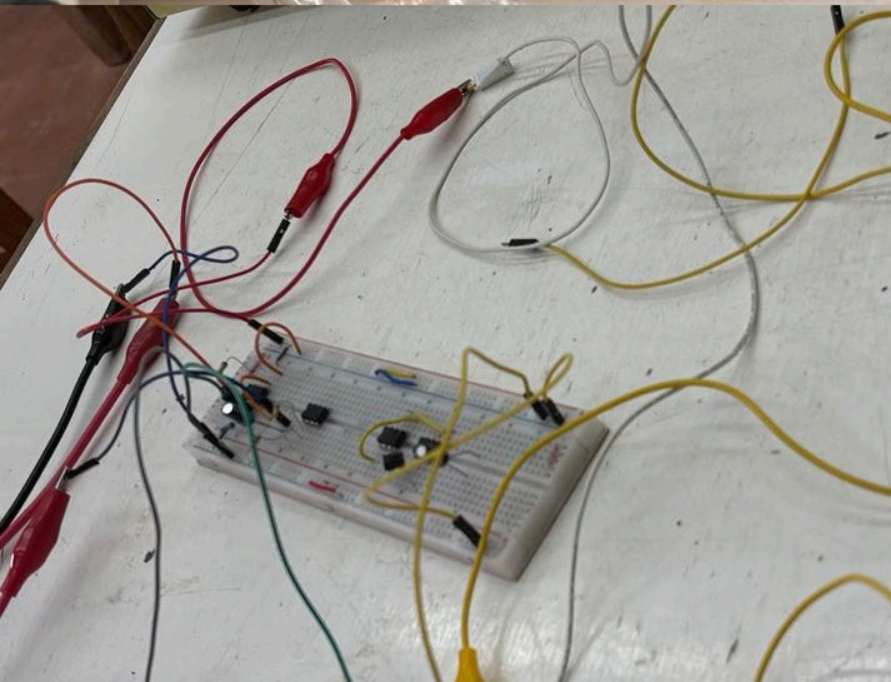
Filters Used in the System

To ensure clean and meaningful EEG data, we implemented High-Pass Filter (HPF) and Low-Pass Filter (LPF) stages in our system.

The filters helped us to:

- Remove DC offset and slow baseline drift using the HPF
- Eliminate high-frequency noise and muscle artifacts using the LPF
- Isolate the useful EEG frequency range

This step ensured that only relevant brain activity was passed to the next stage.

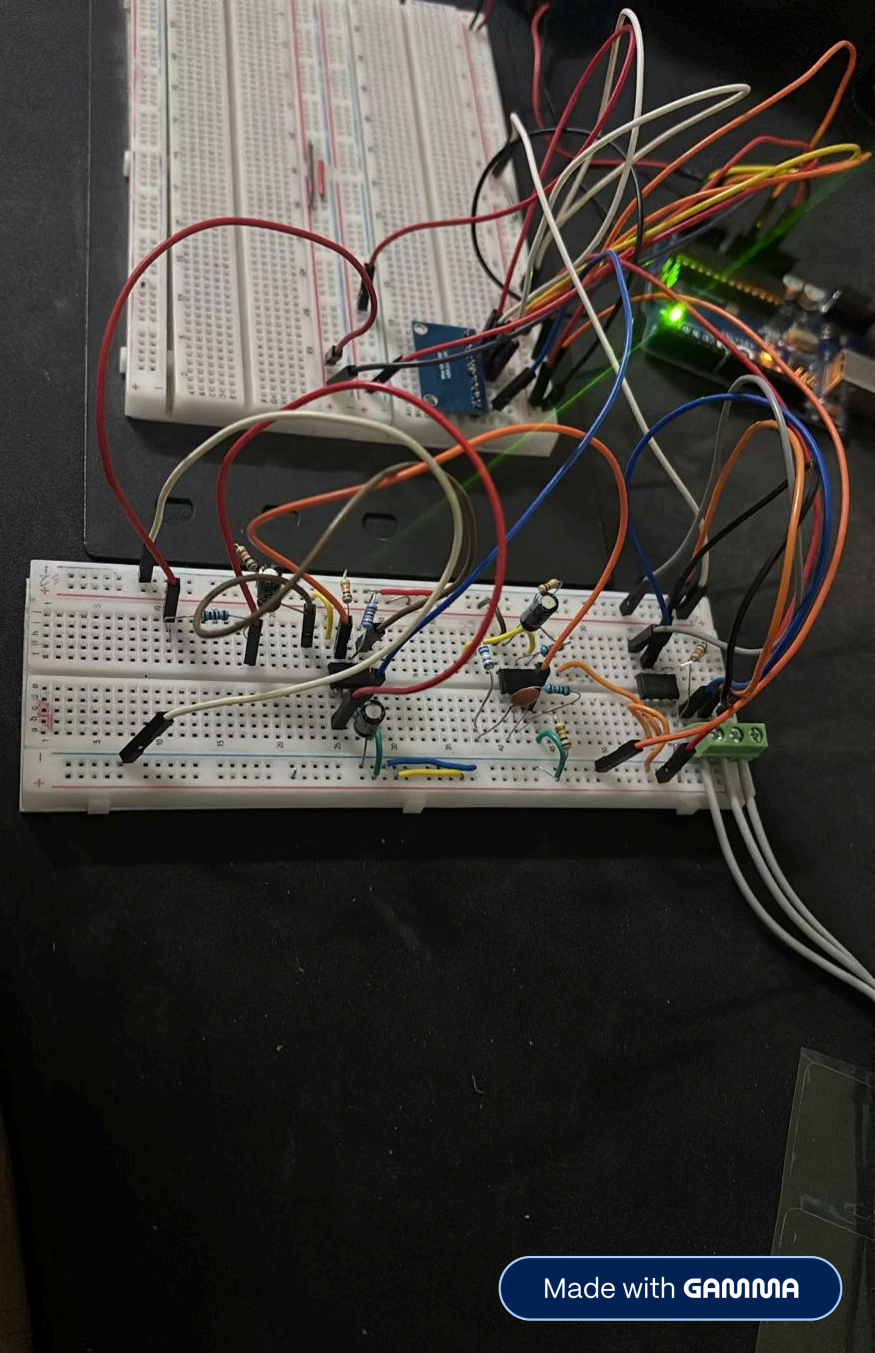


Hardware Implementation: Building the Core Circuit

Our EEG acquisition hardware design focused on robust signal processing through key components:

- **Amplifier:** Handles low-amplitude EEG signals, rejecting common-mode noise for better signal-to-noise ratio.
- **Band-Pass Filtering:** Selectively passes 0.2 - 25 Hz frequencies, eliminating DC offset and high-frequency interference.
- **Noise Reduction:** Shielding and grounding minimized environmental and physiological noise.

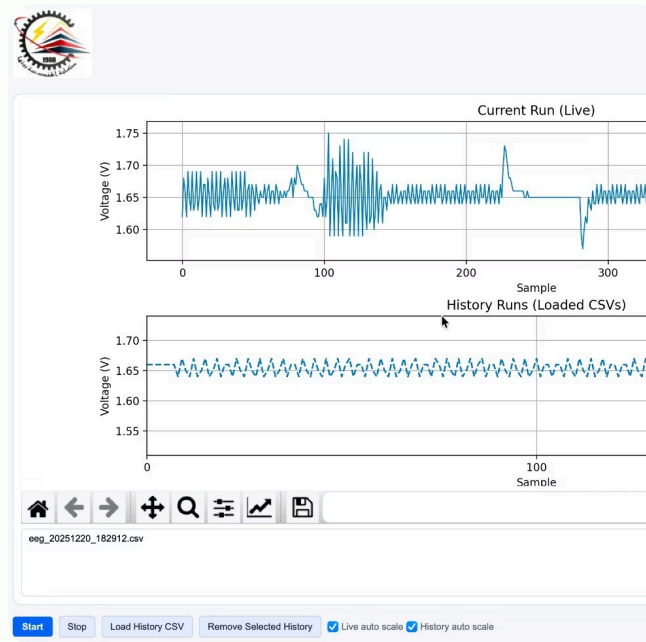
This circuit's successful implementation demonstrates our engineering capability with microvolt-level biomedical signals.



EEG Signal Acquisition: Capturing Brain Activity

Signal acquisition involved key steps:

- **Controlled Conditions:** Recordings minimized external artifacts and ensured data quality.
- **Data Storage:** Signals were digitized and stored for processing and analysis.
- **Reliability Assurance:** Multiple recordings across subjects ensured robust, reliable data.



A1								
	A	B	C	D	E	F	G	H
33741	9.36E+01	#####	#####	#####	2.59E+00	2.59E+00	2.59E+00	2.59E+00
33742	3.91E+02	#####	#####	#####	2.54E+00	2.54E+00	2.54E+00	2.54E+00
33743	7.02E+02	#####	3.36E+02	#####	2.49E+00	2.49E+00	2.49E+00	2.49E+00
33744	8.77E+02	#####	1.14E+03	#####	2.44E+00	2.44E+00	2.44E+00	2.44E+00
33745	9.66E+02	2.55E+02	1.69E+03	1.41E+02	2.39E+00	2.39E+00	2.39E+00	2.39E+00
33746	1.16E+03	2.41E+02	1.92E+03	3.79E+02	2.34E+00	2.34E+00	2.34E+00	2.34E+00
33747	1.50E+03	#####	1.83E+03	6.26E+02	2.29E+00	2.29E+00	2.29E+00	2.29E+00
33748	1.83E+03	#####	1.60E+03	7.95E+02	2.24E+00	2.24E+00	2.24E+00	2.24E+00
33749	1.81E+03	#####	1.51E+03	7.94E+02	2.18E+00	2.18E+00	2.18E+00	2.18E+00
33750	1.23E+03	4.99E+01	1.52E+03	6.03E+02	2.13E+00	2.13E+00	2.13E+00	2.13E+00
33751	6.01E+02	2.68E+02	1.28E+03	3.56E+02	2.08E+00	2.08E+00	2.08E+00	2.08E+00
33752	5.39E+02	1.47E+02	8.17E+02	1.98E+02	2.03E+00	2.03E+00	2.03E+00	2.03E+00
33753	7.52E+02	#####	4.60E+02	1.58E+02	1.98E+00	1.98E+00	1.98E+00	1.98E+00
33754	6.39E+02	#####	3.14E+02	1.84E+02	1.92E+00	1.92E+00	1.92E+00	1.92E+00
33755	2.91E+02	2.85E+01	2.42E+02	1.81E+02	1.87E+00	1.87E+00	1.87E+00	1.87E+00
33756	1.05E+02	3.57E+02	2.40E+02	1.28E+02	1.82E+00	1.82E+00	1.82E+00	1.82E+00
33757	4.51E+01	3.41E+02	1.29E+02	8.93E+01	1.77E+00	1.77E+00	1.77E+00	1.77E+00
33758	#####	1.21E+02	#####	6.02E+01	1.72E+00	1.72E+00	1.72E+00	1.72E+00
33759	#####	3.34E+01	#####	9.88E-01	1.66E+00	1.66E+00	1.66E+00	1.66E+00
33760	#####	1.79E+02	#####	#####	1.61E+00	1.61E+00	1.61E+00	1.61E+00
33761	#####	3.42E+02	#####	8.65E+00	1.56E+00	1.56E+00	1.56E+00	1.56E+00
33762	#####	1.86E+02	4.30E+02	5.85E+01	1.51E+00	1.51E+00	1.51E+00	1.51E+00
33763	1.83E+02	#####	3.44E+02	6.16E+01	1.46E+00	1.46E+00	1.46E+00	1.46E+00
33764	1.73E+02	#####	#####	#####	1.41E+00	1.41E+00	1.41E+00	1.41E+00
33765	#####	#####	#####	#####	1.36E+00	1.36E+00	1.36E+00	1.36E+00

Project Testing: Validating System Performance

Rigorous testing ensures our EEG system's reliability and accuracy, crucial for clinical and research applications.

Module-Level Testing

Individual components (amplifiers, filters) are tested for electrical characteristics and signal integrity.

System Integration Test

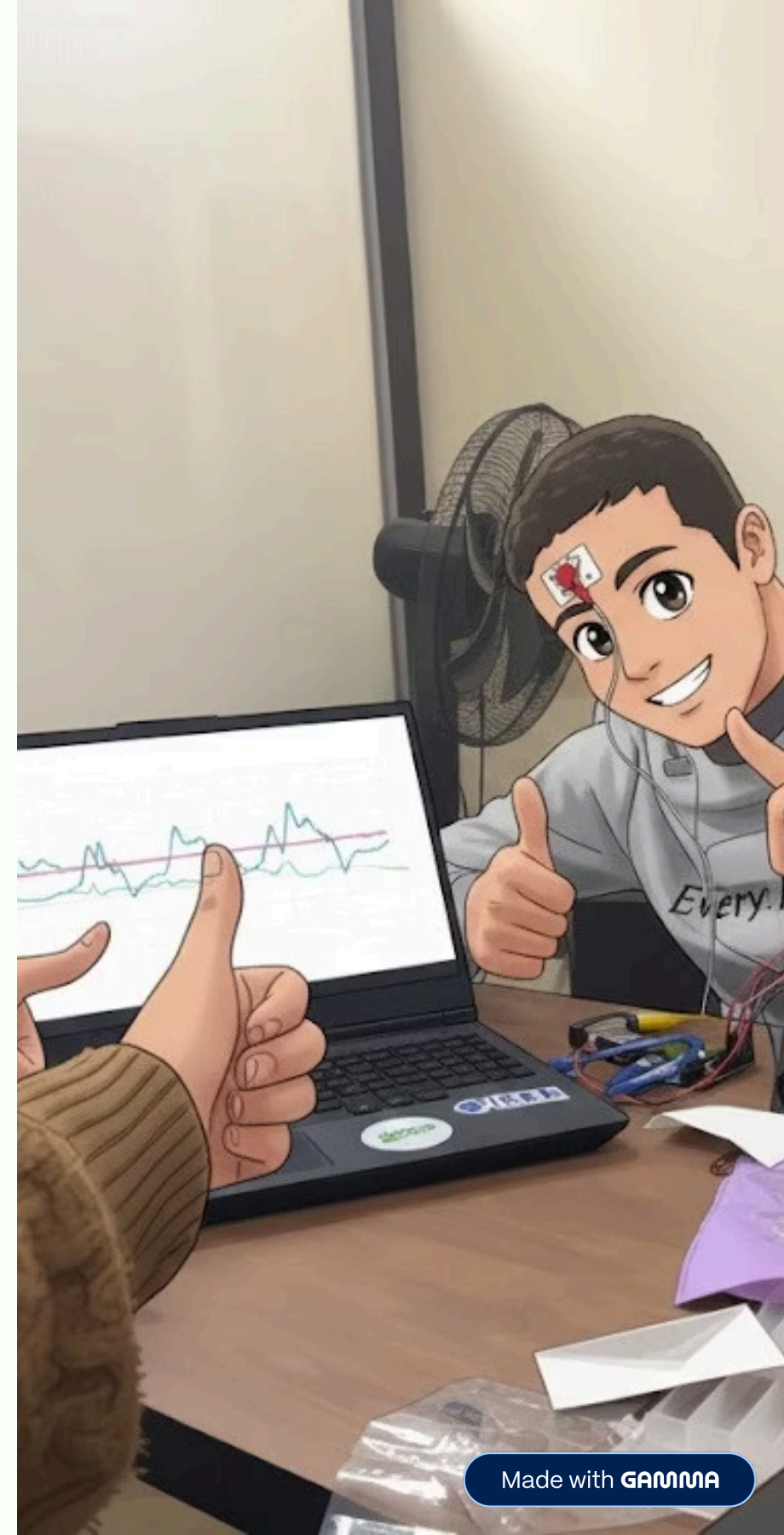
The assembled system is evaluated for seamless component interaction and data flow.

Performance Validation

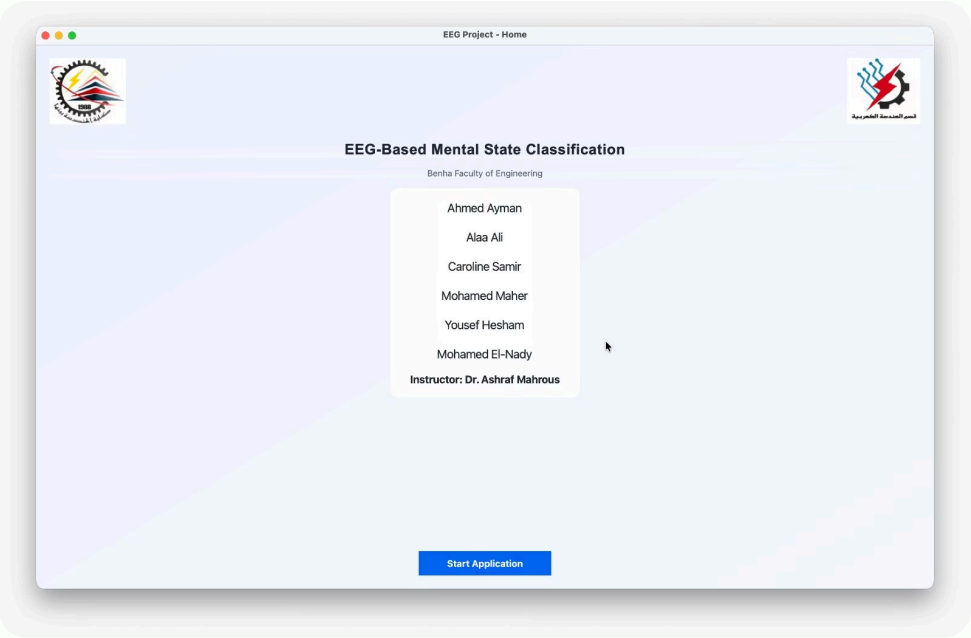
Comparative analysis with commercial EEG systems using simulated and real brain signals.

User Acceptance Testing

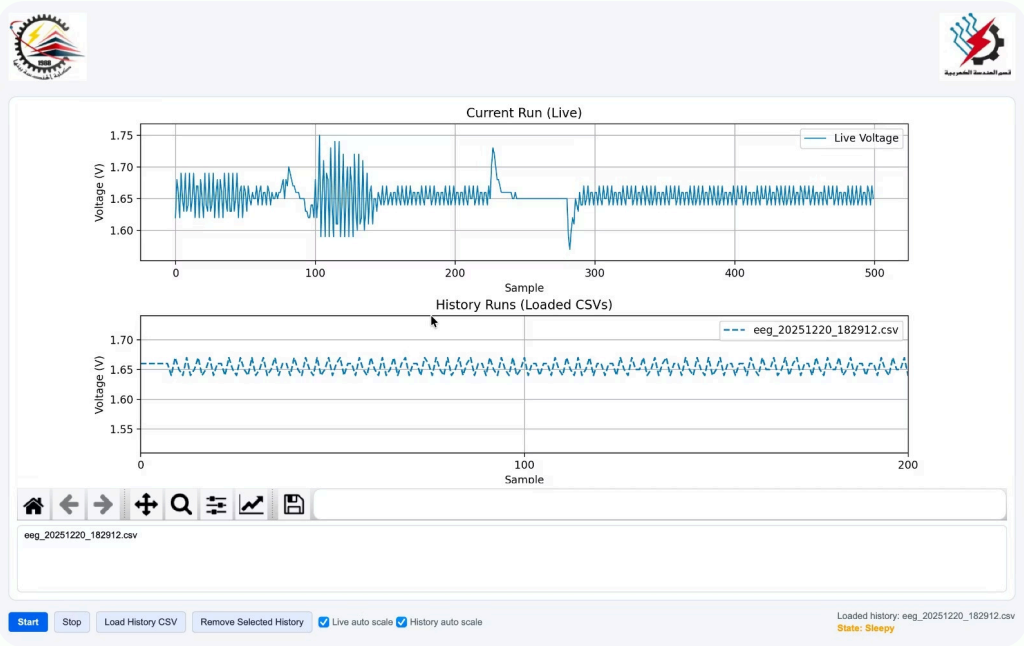
Feedback from domain experts and potential users informs final adjustments and usability.



User interface



Start Window



Signal Visualization

Encountered Difficulties

- ▼ **Lack of Clear References for Educational EEG Circuits**
- ▼ **Unexpected Initial Measurements (AD620)**
- ▼ **ADC Input Range Limitation:**
- ▼ **Difficulty in achieving stable electrode contact**